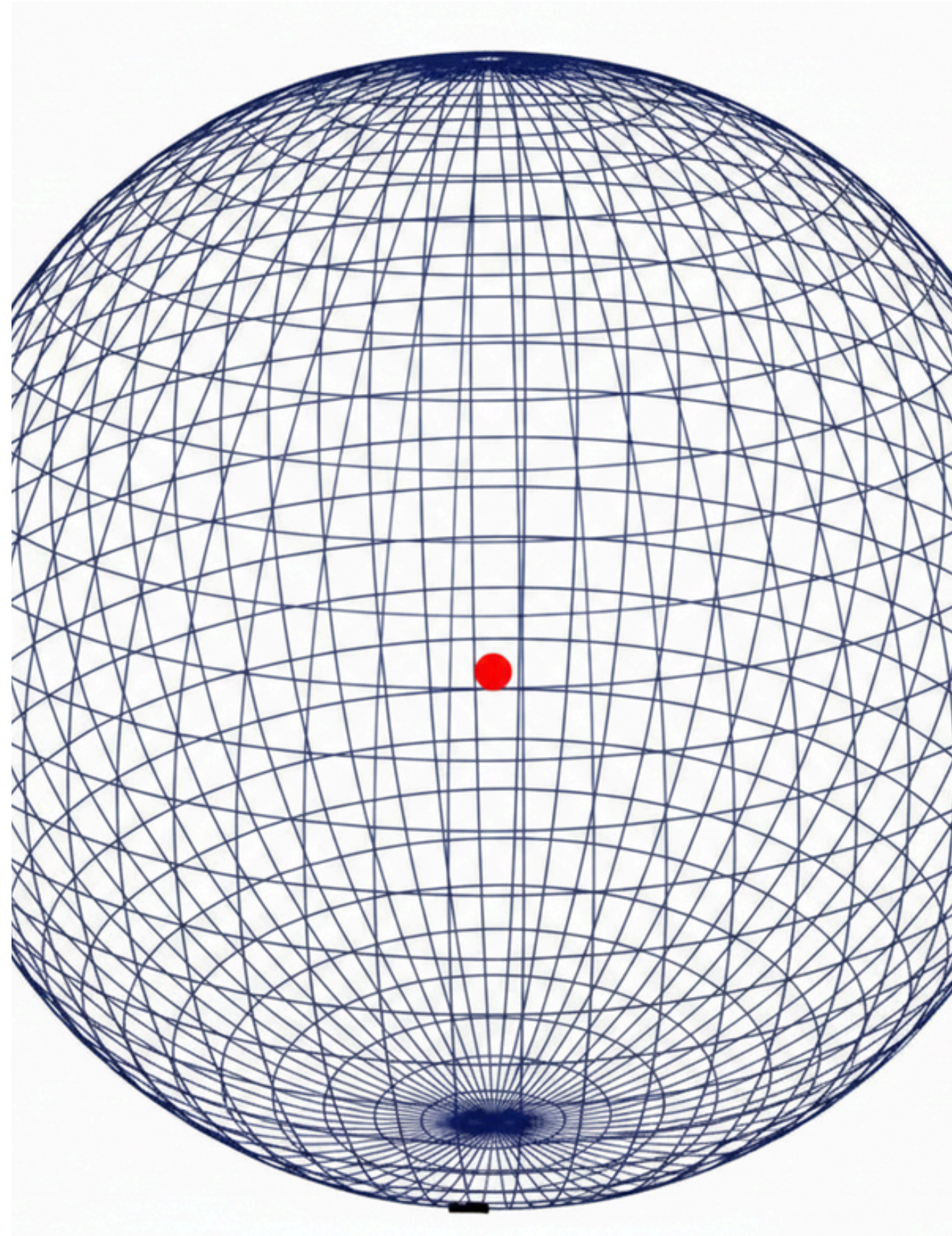


Wi-Fi Indoor Localization with Machine Learning

Mattia Coriale (S5640317) & Gabriele Alessandria (S5622102)





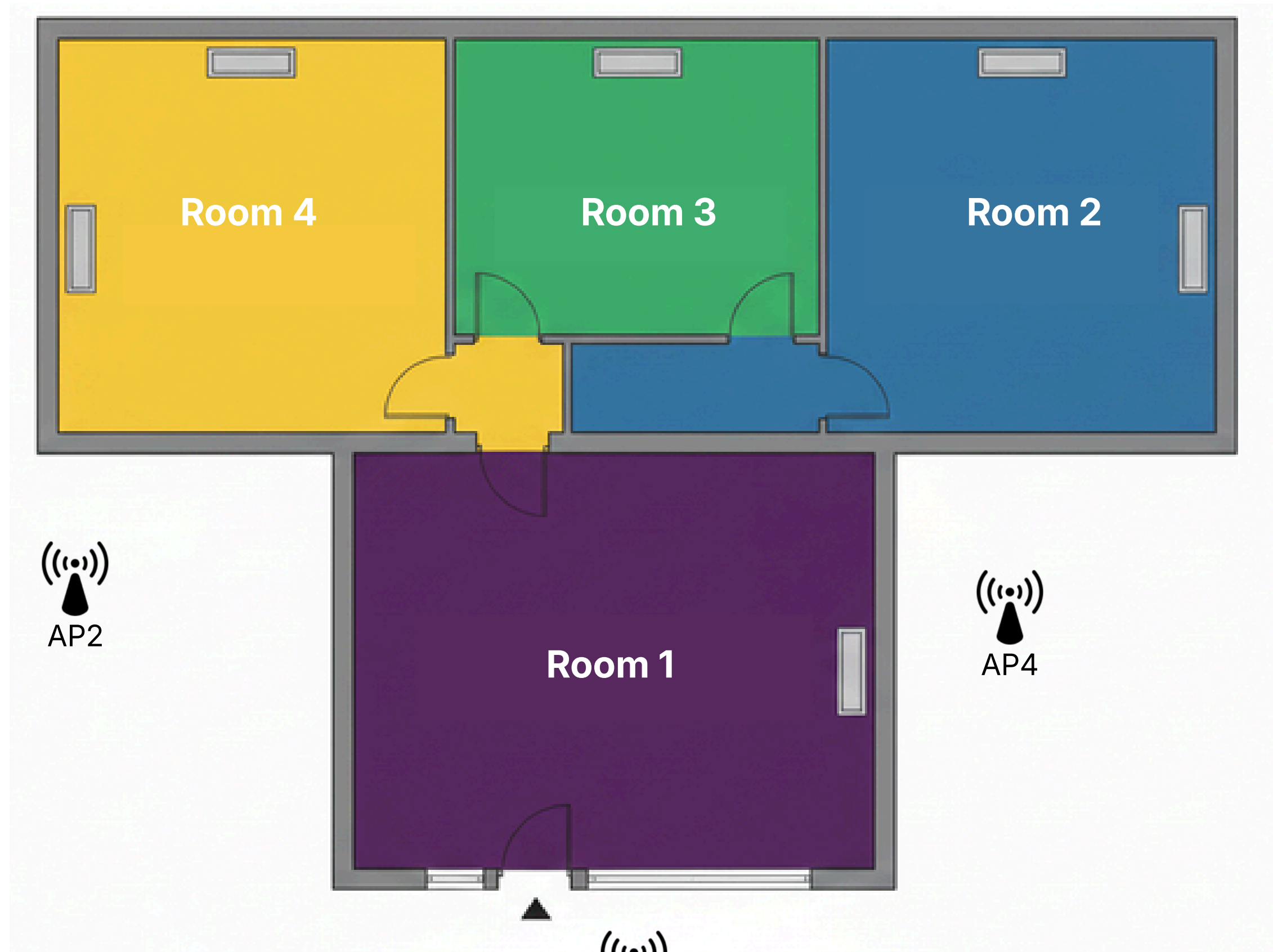
Abstract

- Project Overview
- Dataset
- Implementation
- Results
 - Confusion Matrix Analysis
 - Accuracy Assessment
 - Additional evaluation metrics
- Further Analysis
 - Error Analysis: Evaluating the Best-Performing Models
- Conclusions

Project Overview

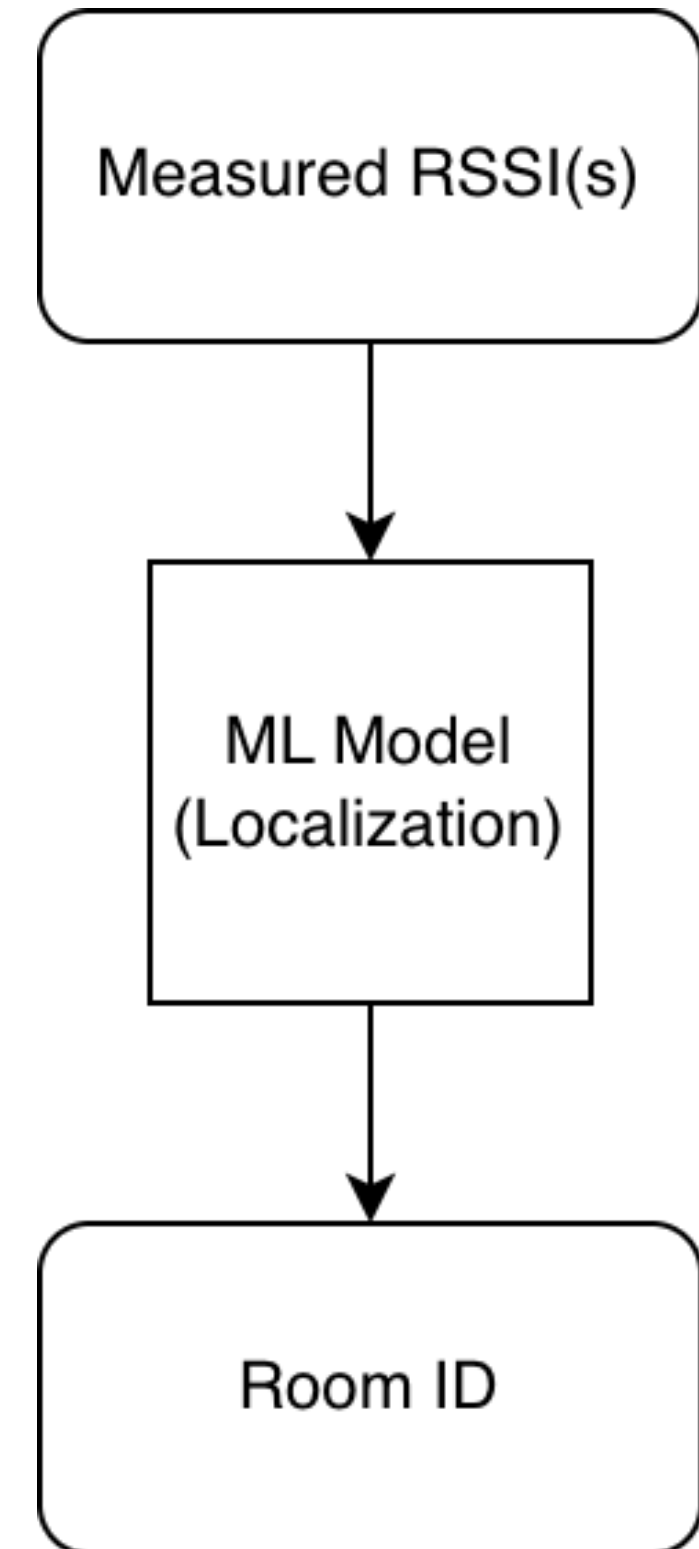
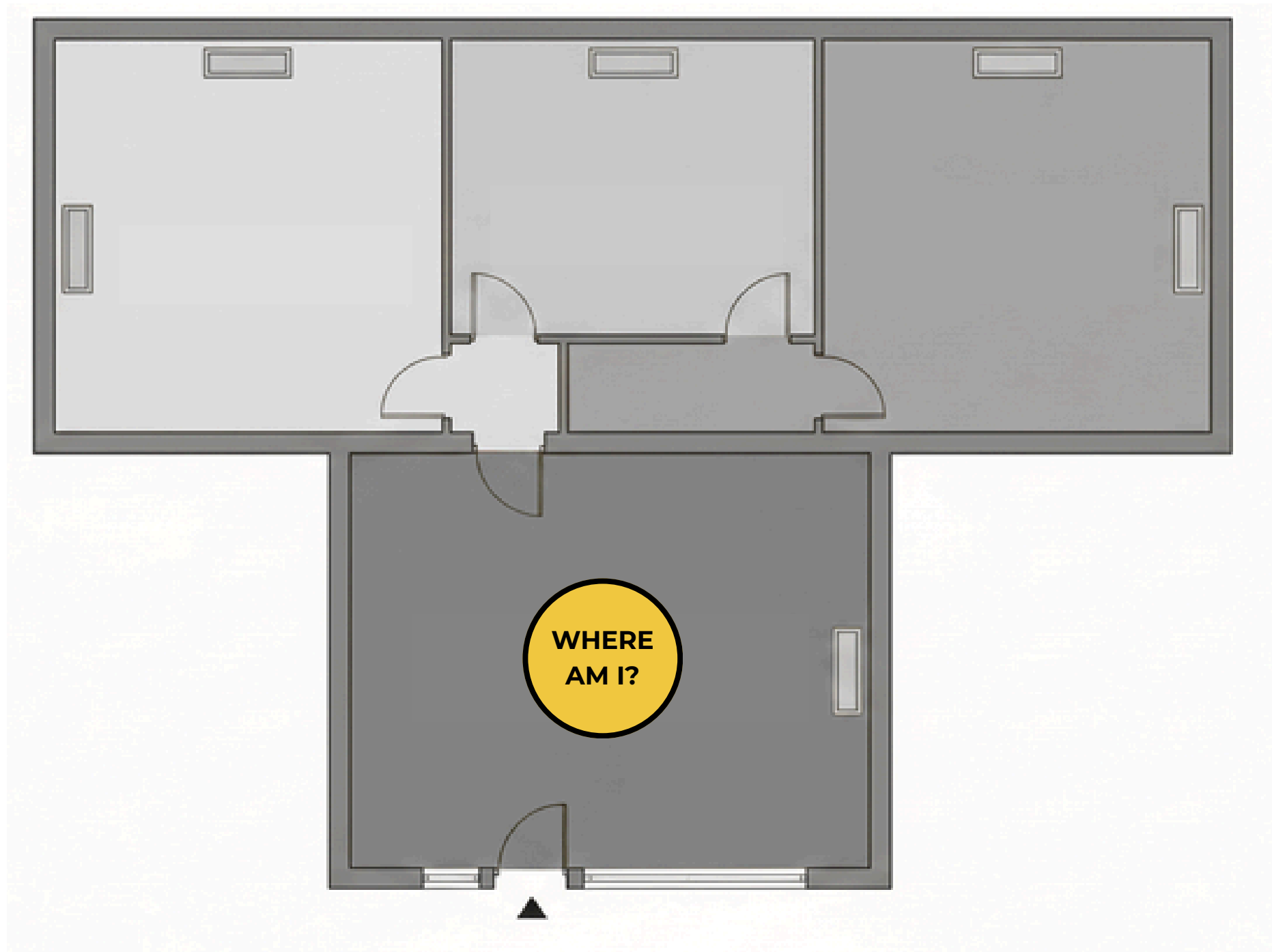


Real dataset room disposition is unknown

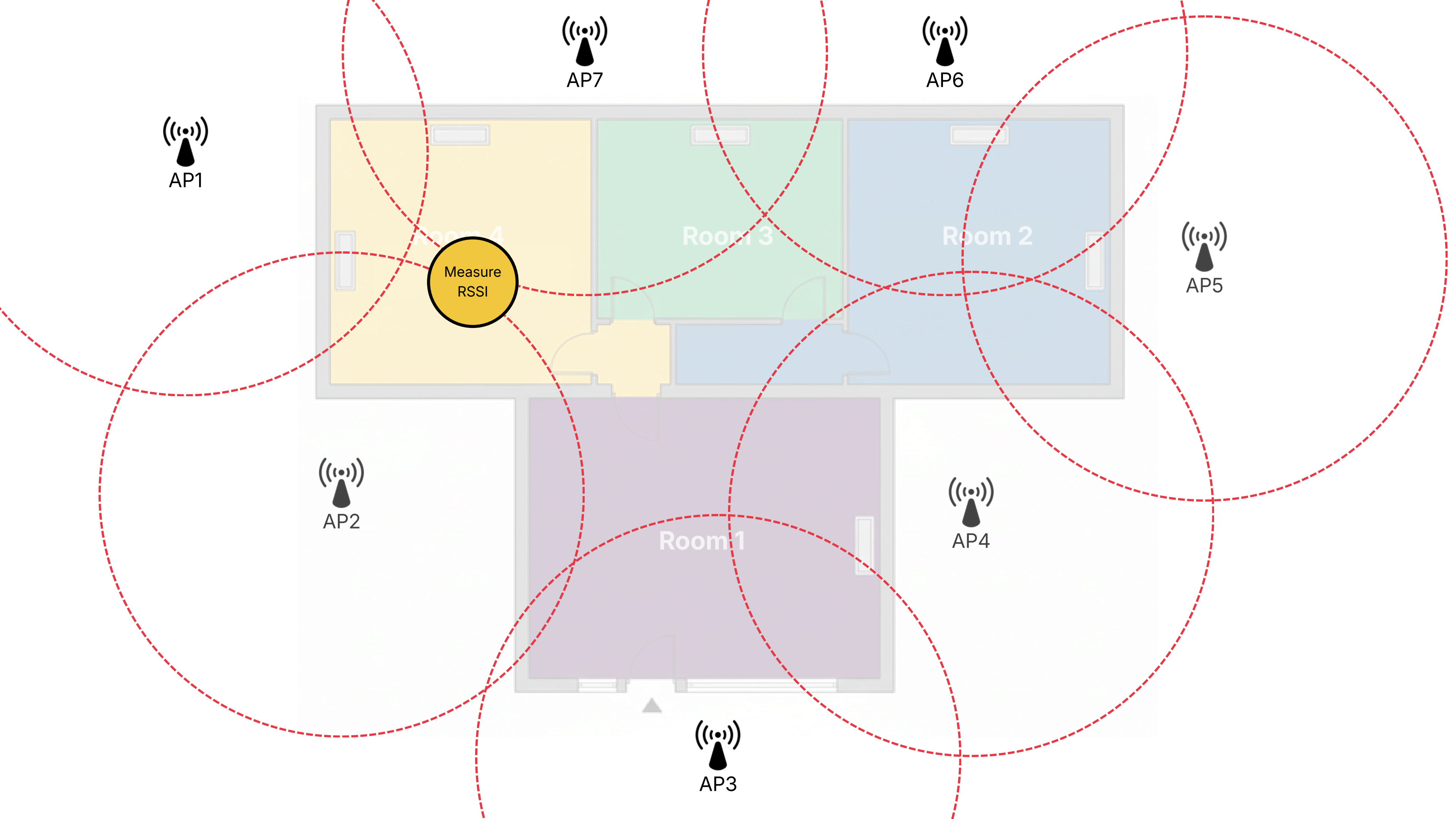


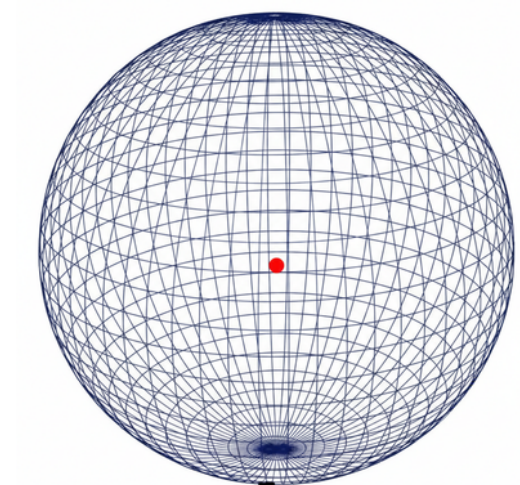
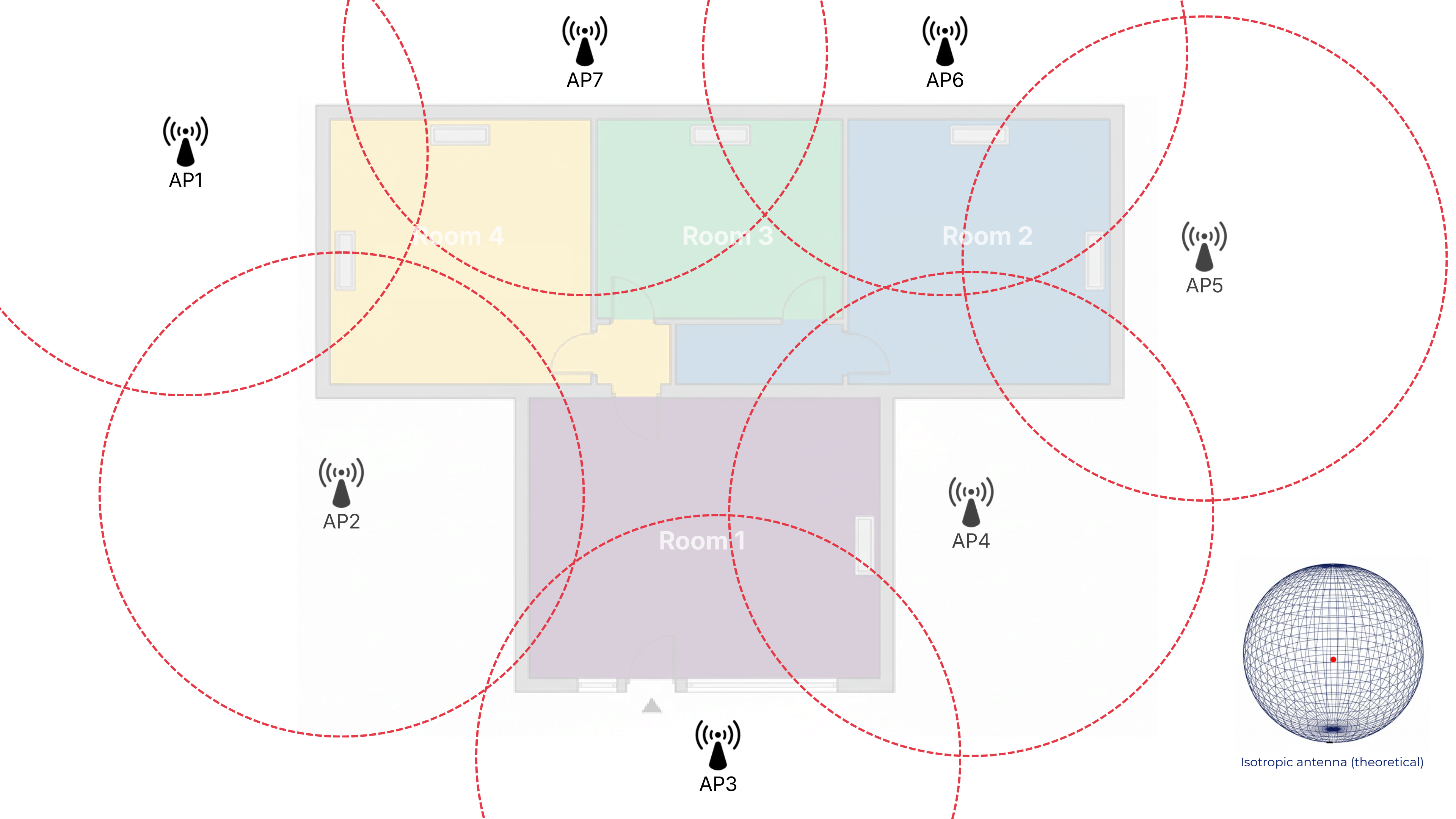
Real dataset router disposition is unknown

The Goal

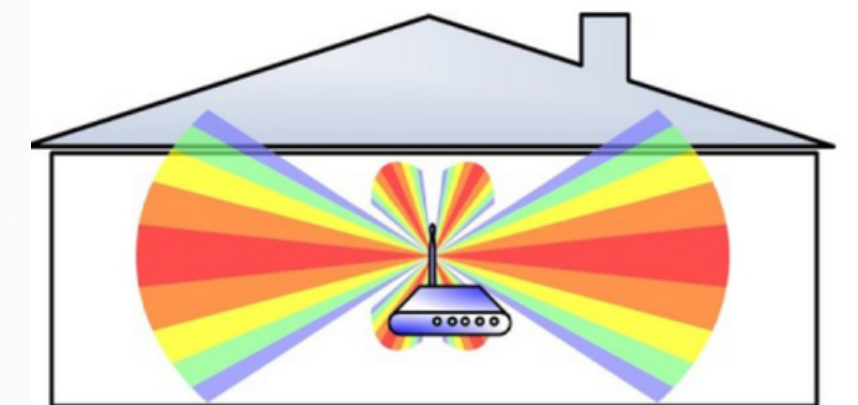
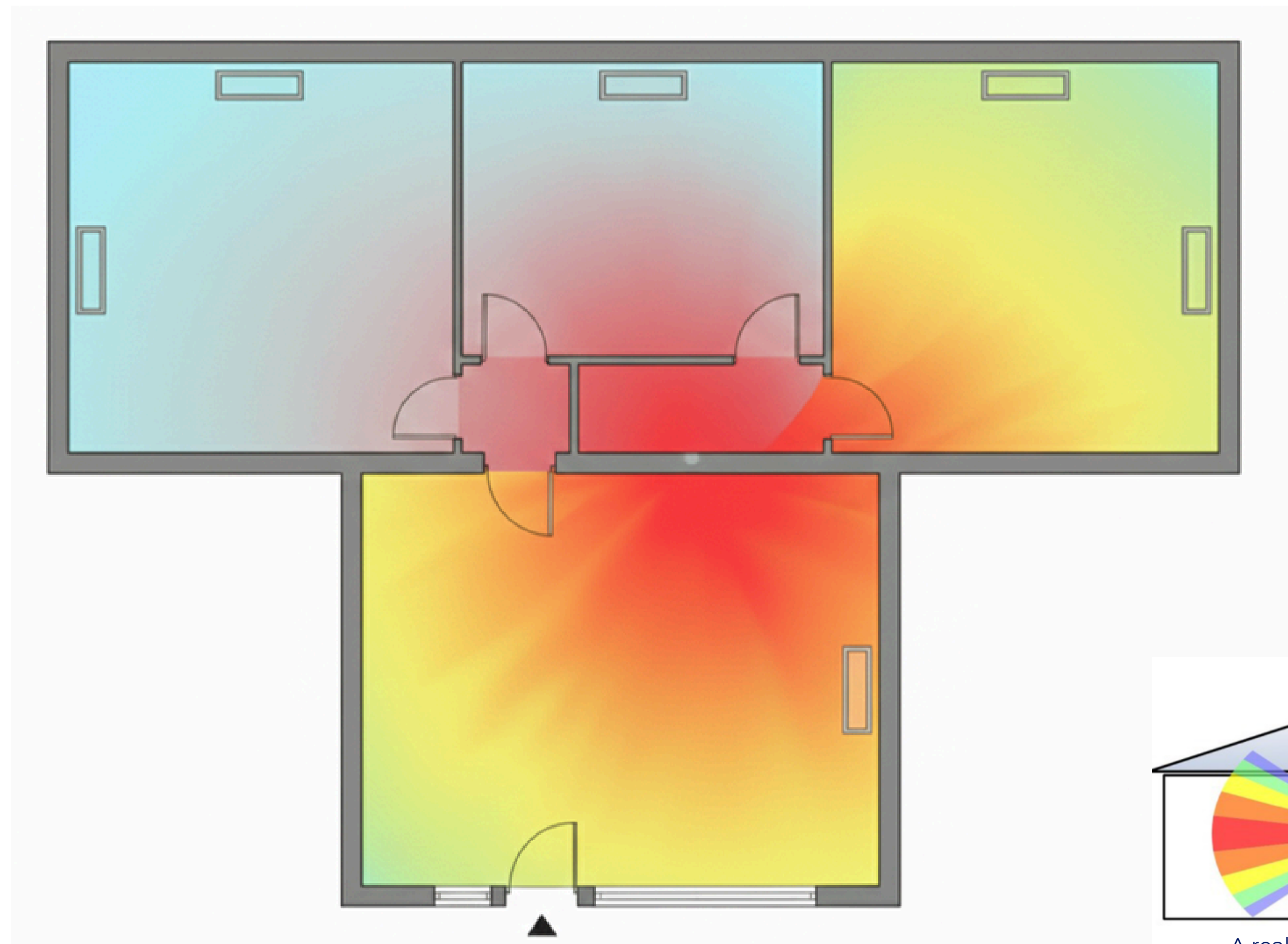


Core Idea





Isotropic antenna (theoretical)

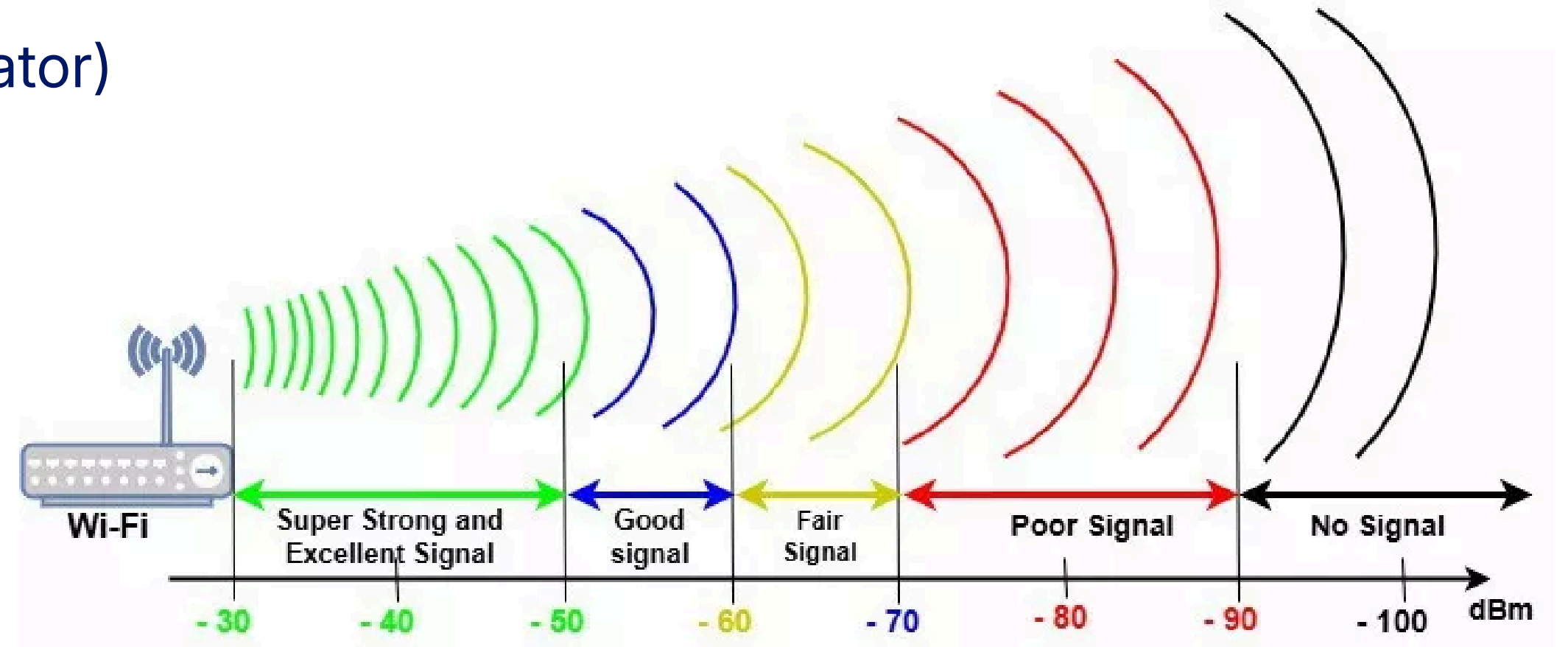


A real antenna is not isotropic

Real dataset overall signal heatmap is unknown

Signal Strength

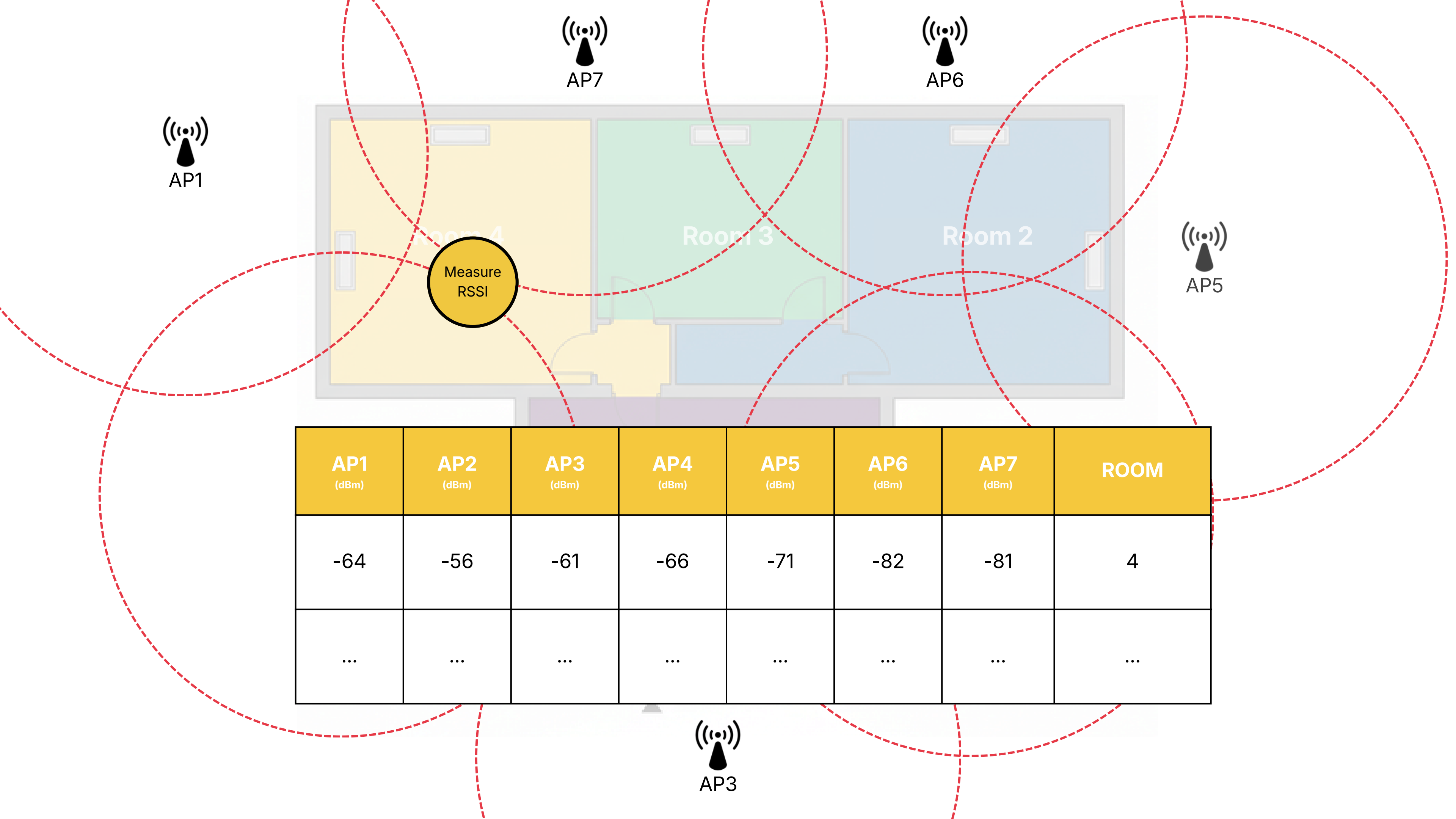
RSSI (Received Signal Strength Indicator)



RSSI decreases with distance from the AP

dBm

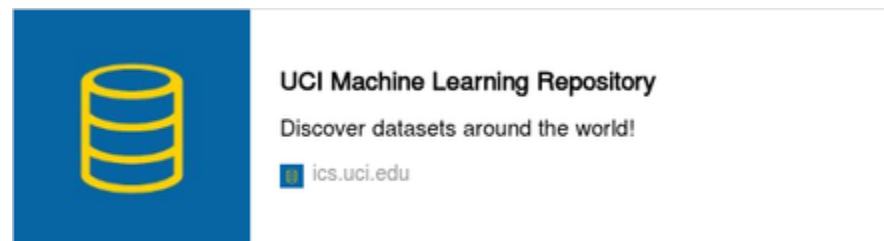
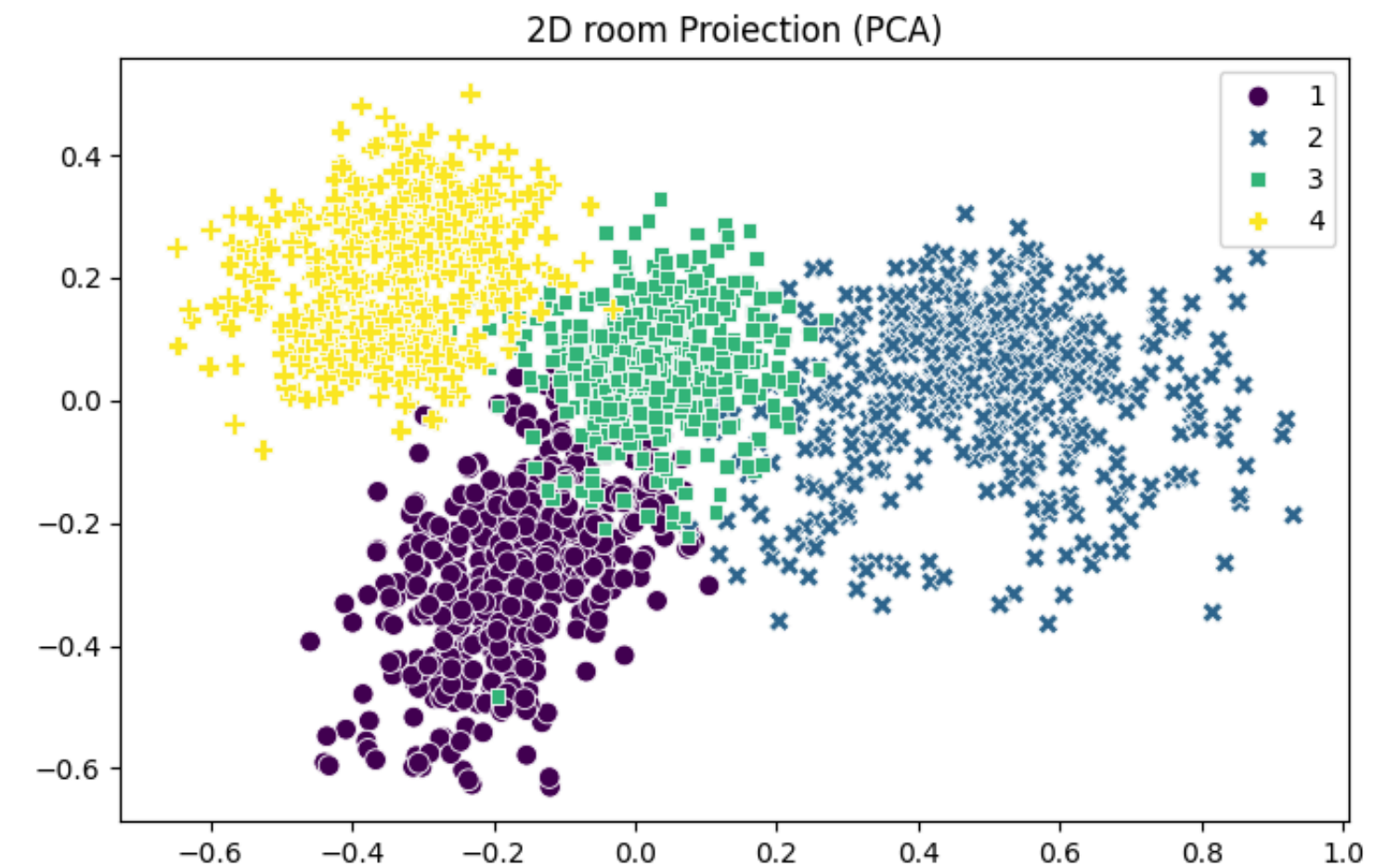
0 dBm = 1 milliwatt (mW)



AP1 (dBm)	AP2 (dBm)	AP3 (dBm)	AP4 (dBm)	AP5 (dBm)	AP6 (dBm)	AP7 (dBm)	ROOM
-64	-56	-61	-66	-71	-82	-81	4
...	...	...	...	...	...	...	...

Dataset

AP1 (dBm)	AP2 (dBm)	AP3 (dBm)	AP4 (dBm)	AP5 (dBm)	AP6 (dBm)	AP7 (dBm)	ROOM
-64	-56	-61	-66	-71	-82	-81	4
...	...	...	...	...	...	...	...



~2000 records

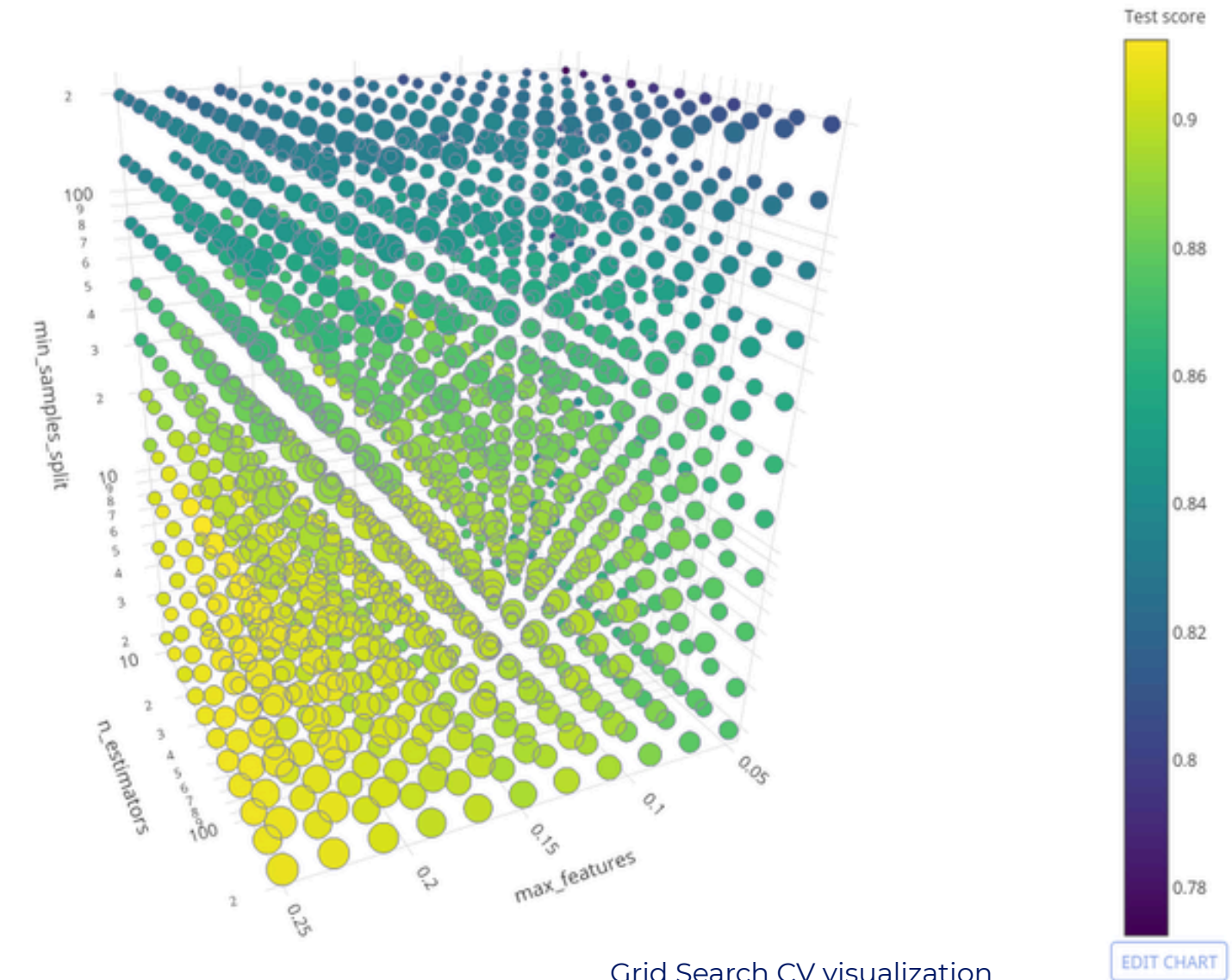
Implementation

GridSearchCV

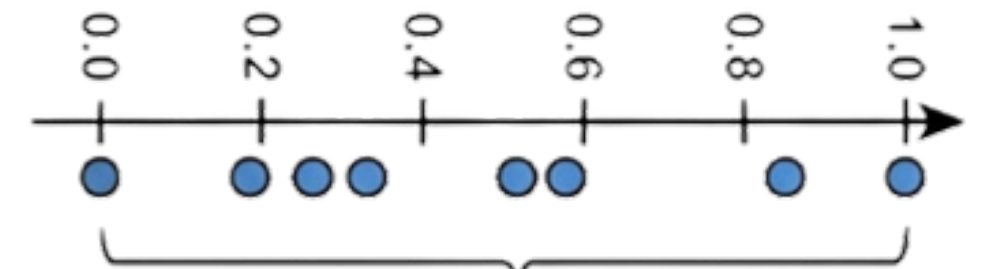
+ MinMax Scaler

```
GridSearchCV

for model_name, config in models_config.items():
    print(f"Training {model_name}...")
    if config['params']:
        clf = GridSearchCV(config['model'], config['params'],
cv=5, scoring='accuracy', n_jobs=-1)
    else:
        clf = config['model']
```

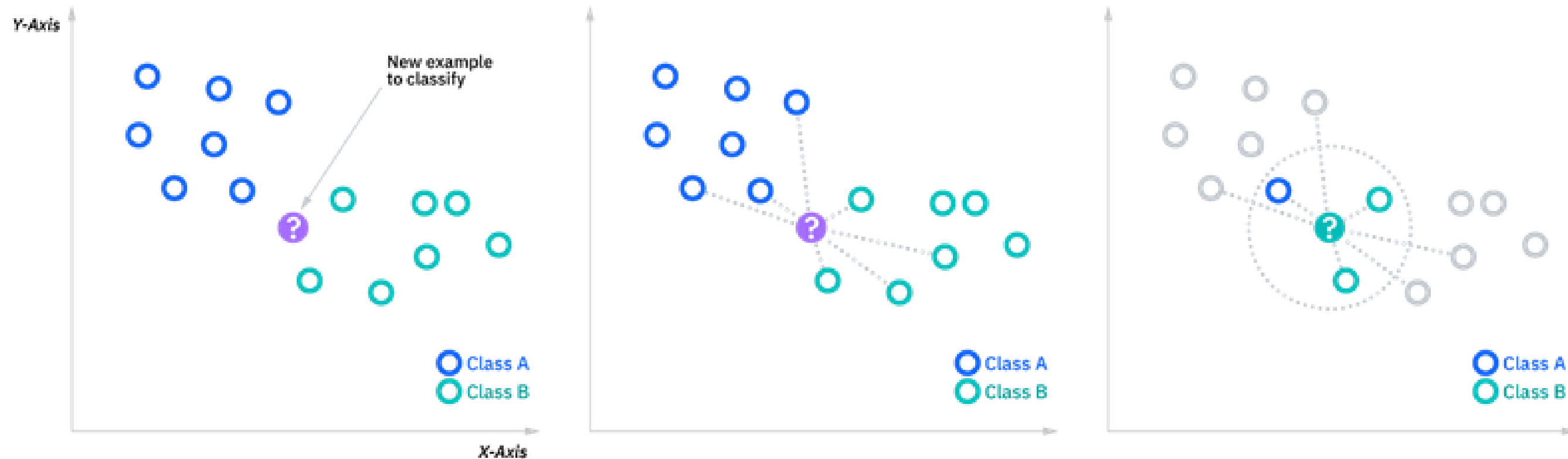


Grid Search CV visualization



Scaled Data
(Normalized Range)

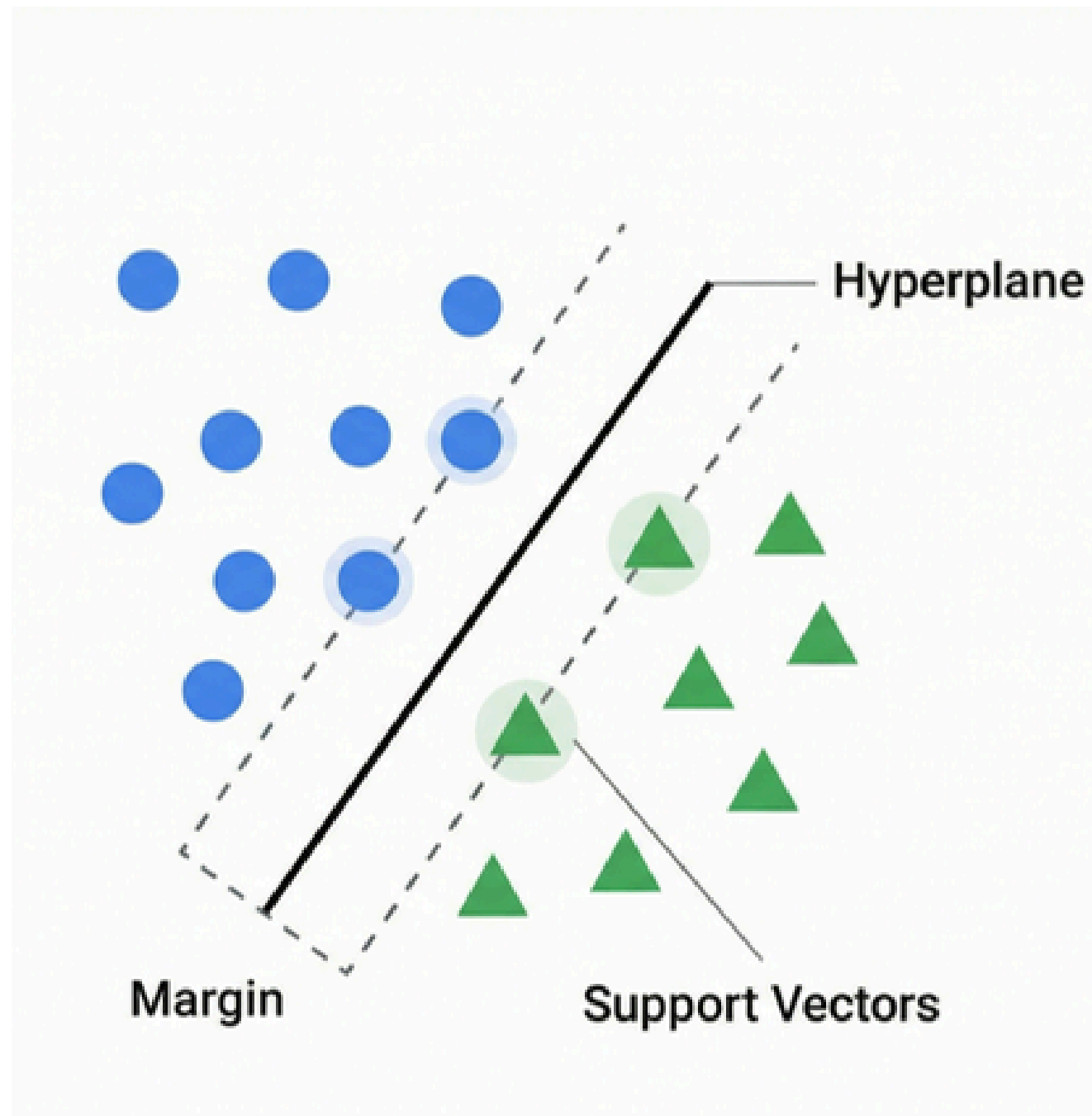
KNN



```
KNeighbors

{'KNN': {
    'model': KNeighborsClassifier(),
    'params': {'n_neighbors': [3, 5, 7, 9], 'weights':
['uniform', 'distance'], 'metric': ['euclidean',
'manhattan']}}
}
```

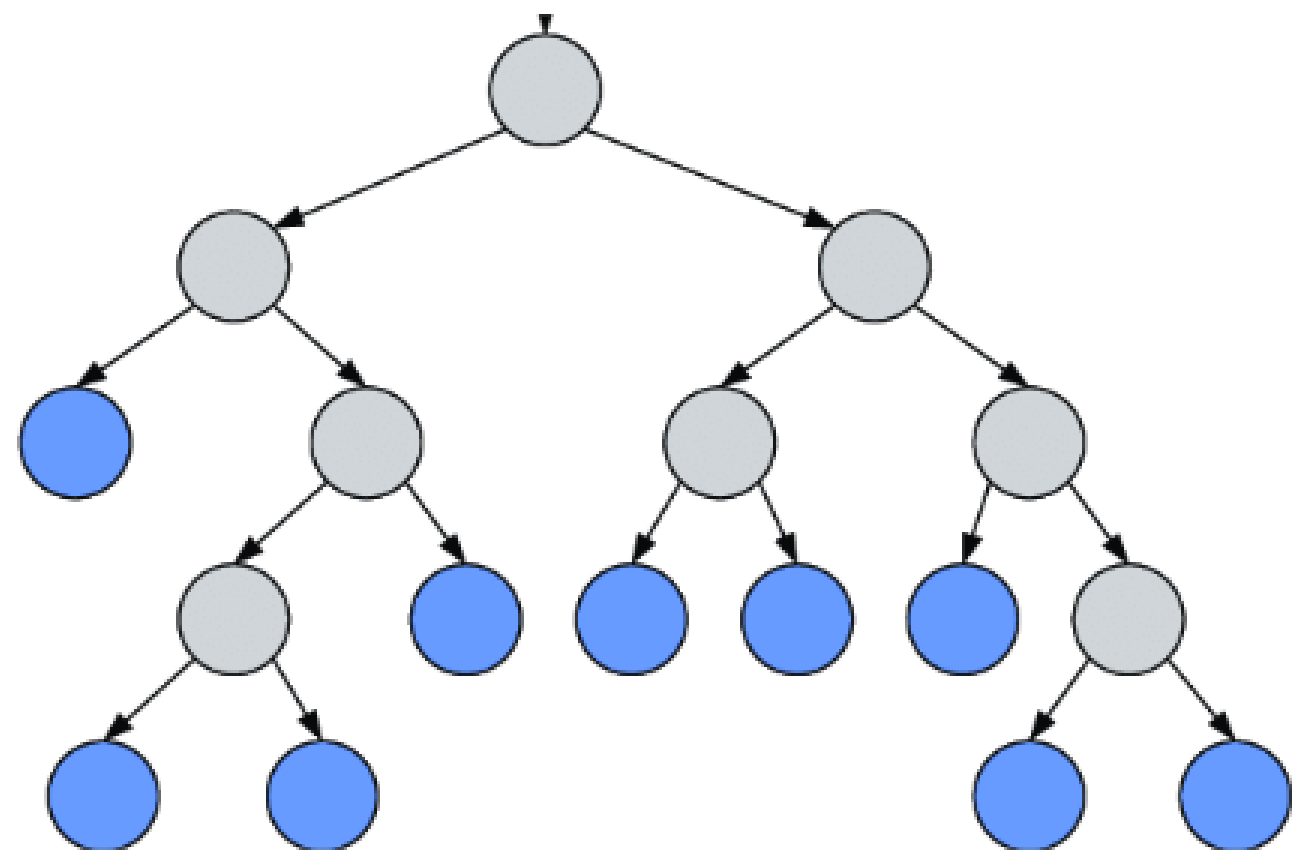
SVC



```
Support Vector Classification

'SVM': {
    'model': SVC(random_state=0, probability=True),
    'params': {'C': [0.1, 1, 10, 100], 'kernel': ['rbf',
'linear'], 'gamma': ['scale', 'auto']}
},
```

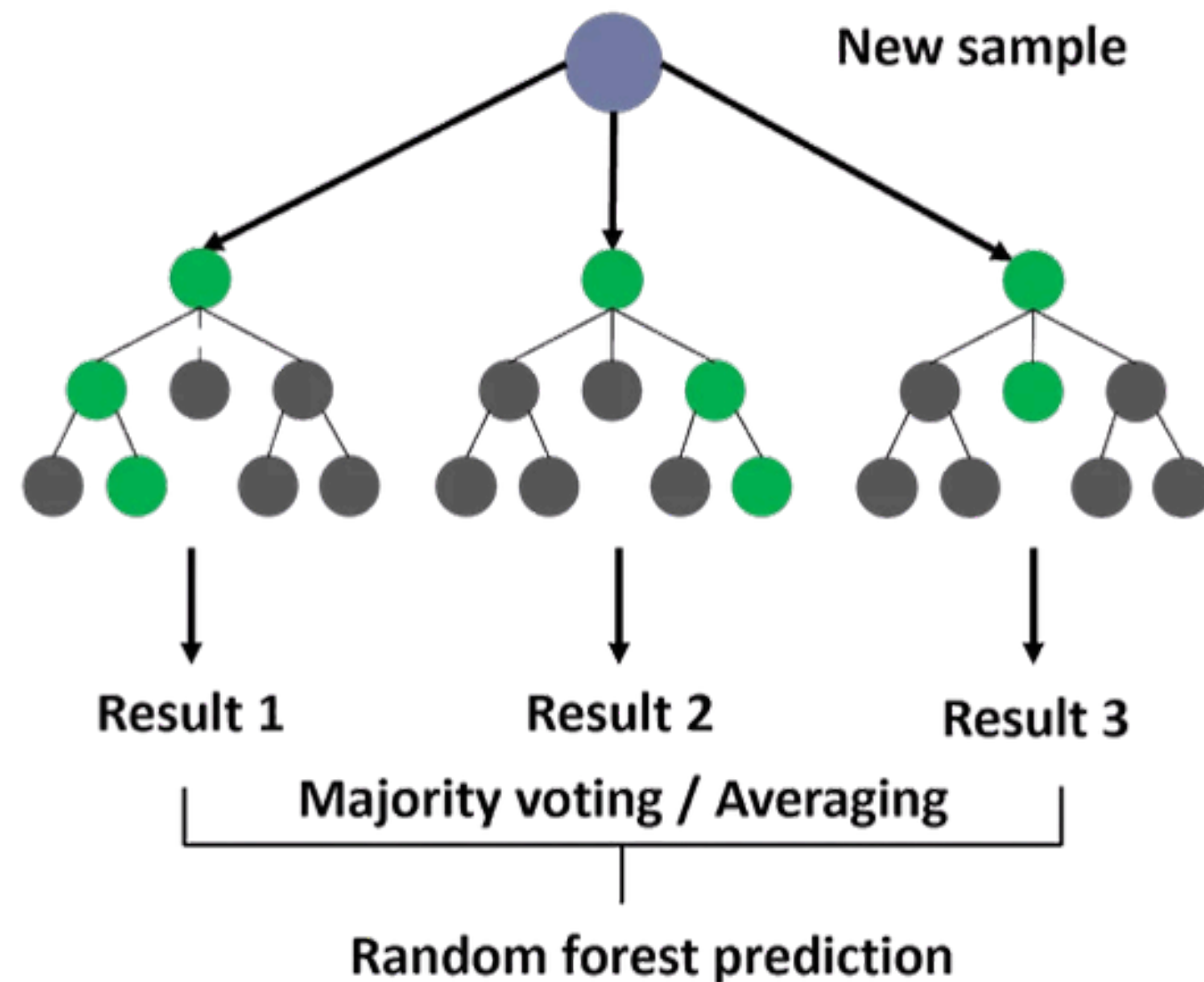
Decision Tree



```
Decision Tree

'Decision Tree': {
    'model': DecisionTreeClassifier(random_state=0),
    'params': {'max_depth': [5, 10, 20, None],
    'criterion': ['gini', 'entropy']}
}
```

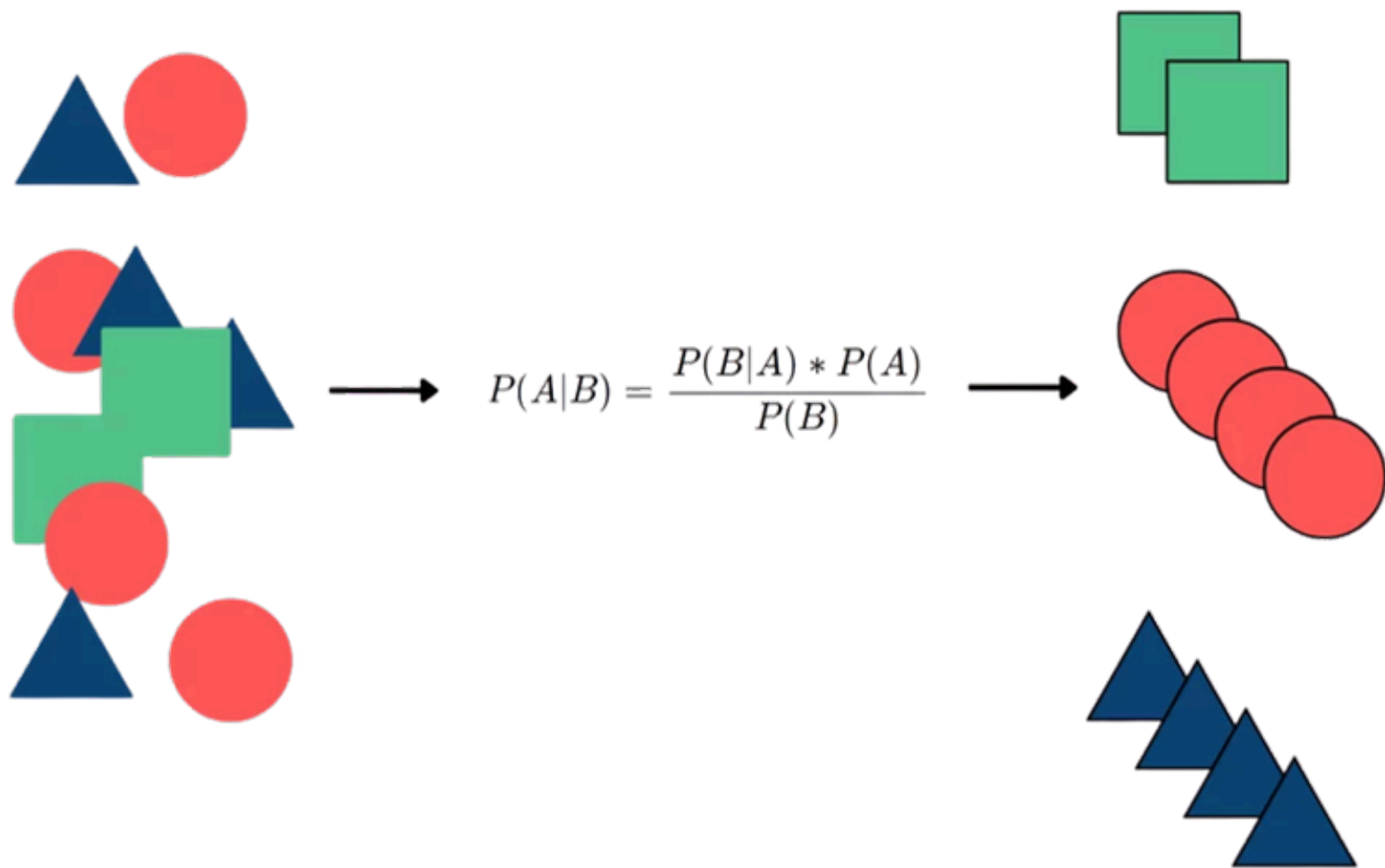
Random Forest



```
Random Forest

'Random Forest': {
    'model': RandomForestClassifier(random_state=0),
    'params': {'n_estimators': [50, 100],
'max_depth': [None, 10, 20]}
},
```

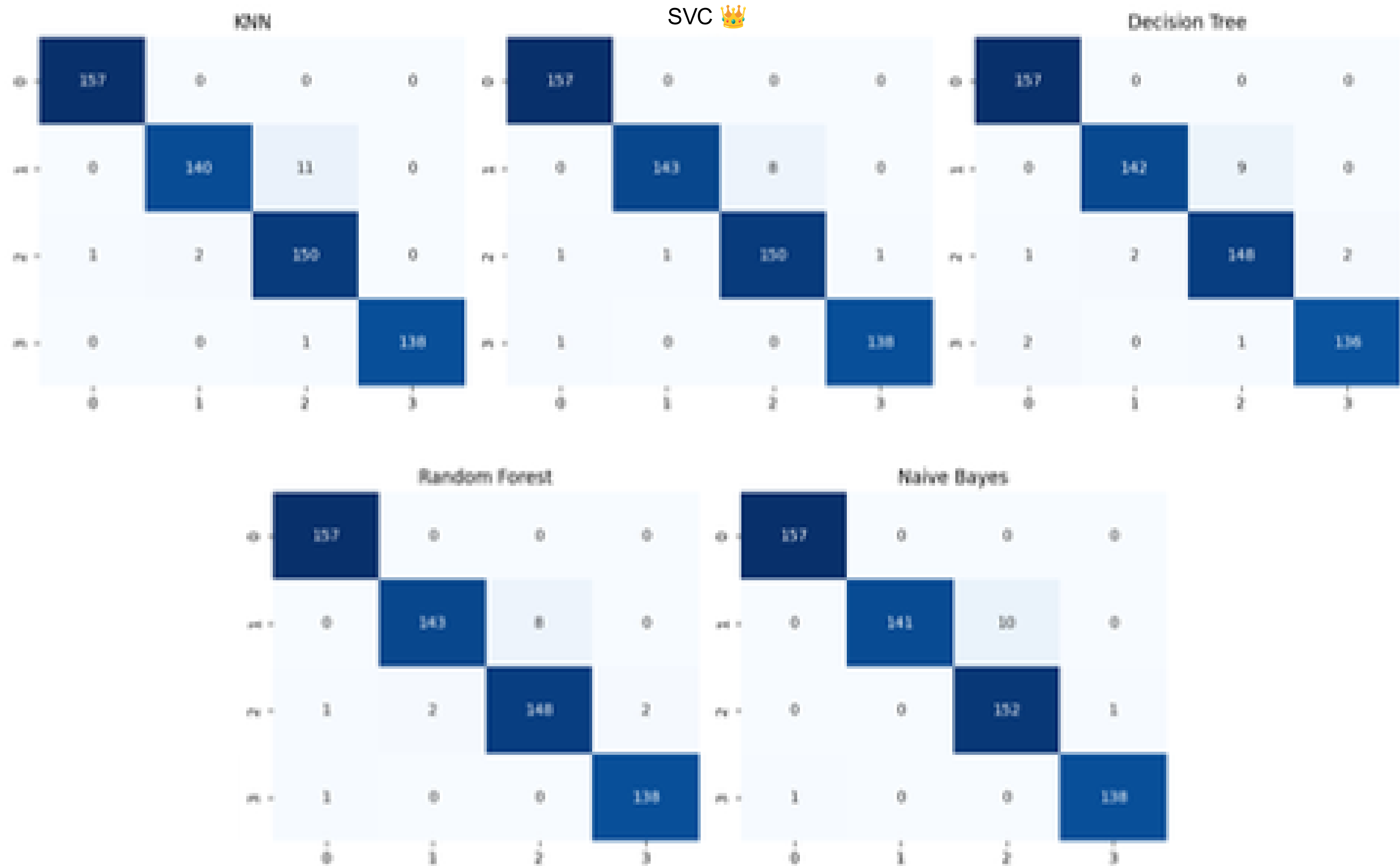

Naive Bayes



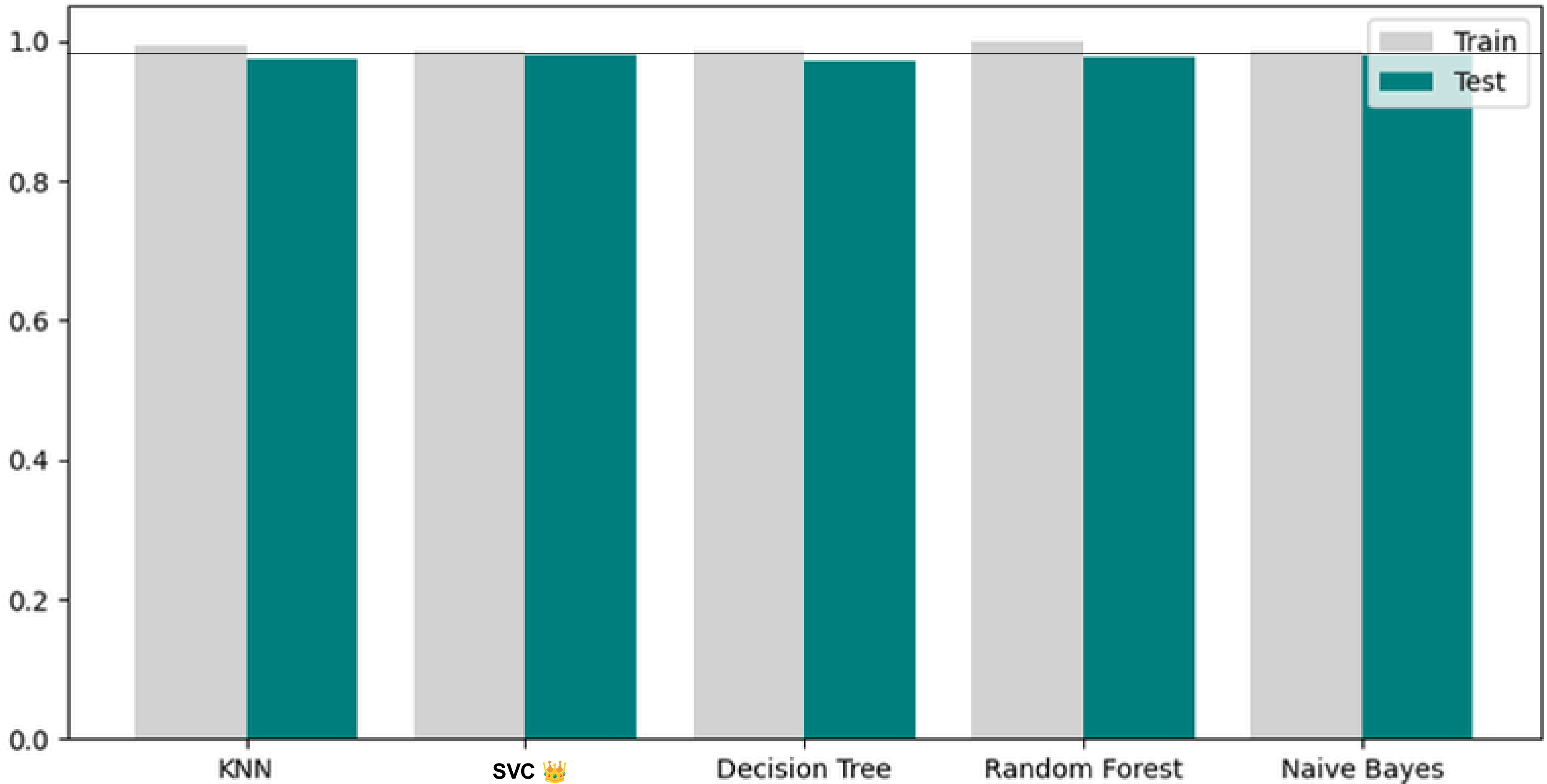
```
Naive Bayes

'Naive Bayes': {
  'model': GaussianNB(),
  'params': {}
}
```


Results



Accuracy Comparison: Train vs Test



Precision Comparison, Recall and F1-Score

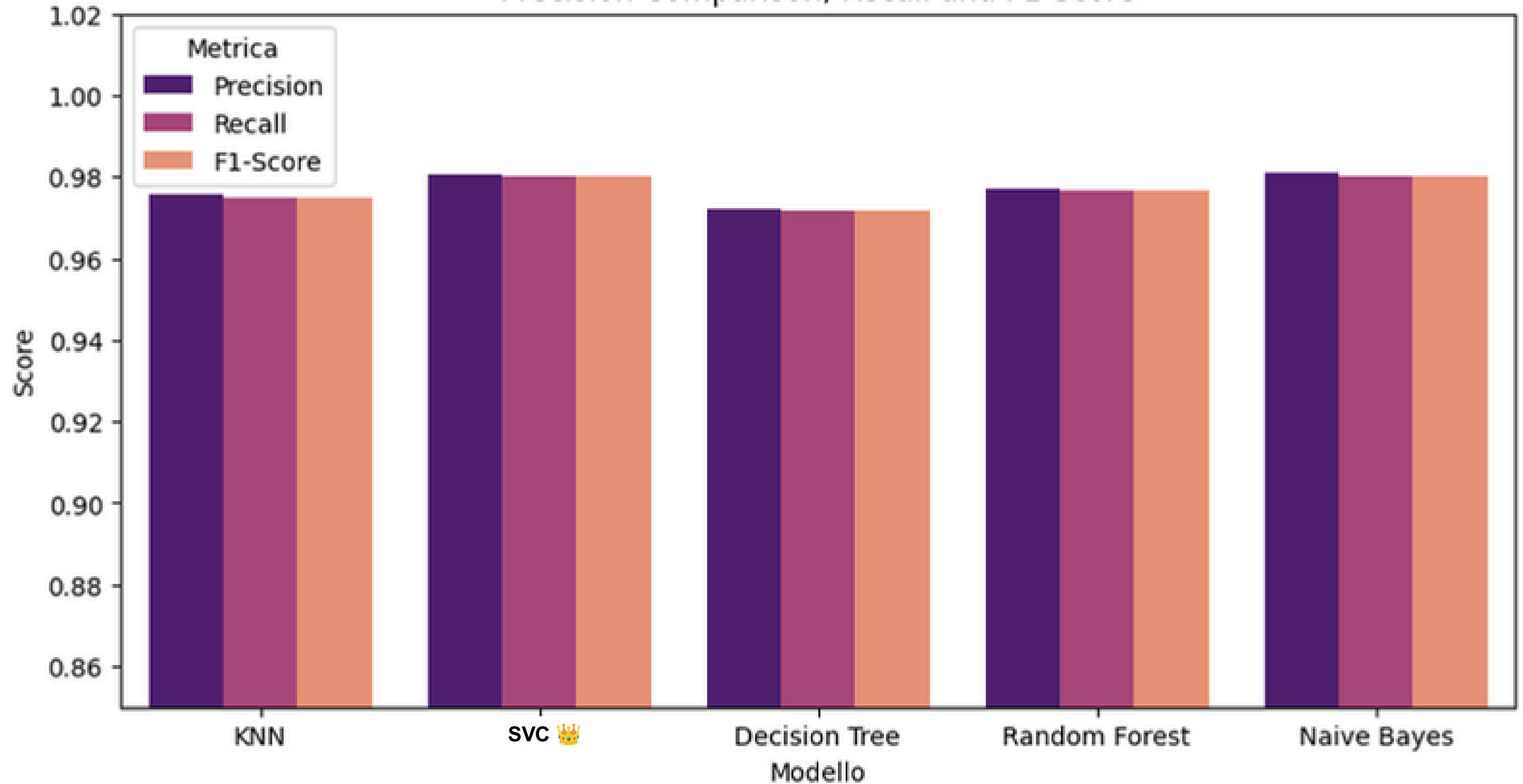
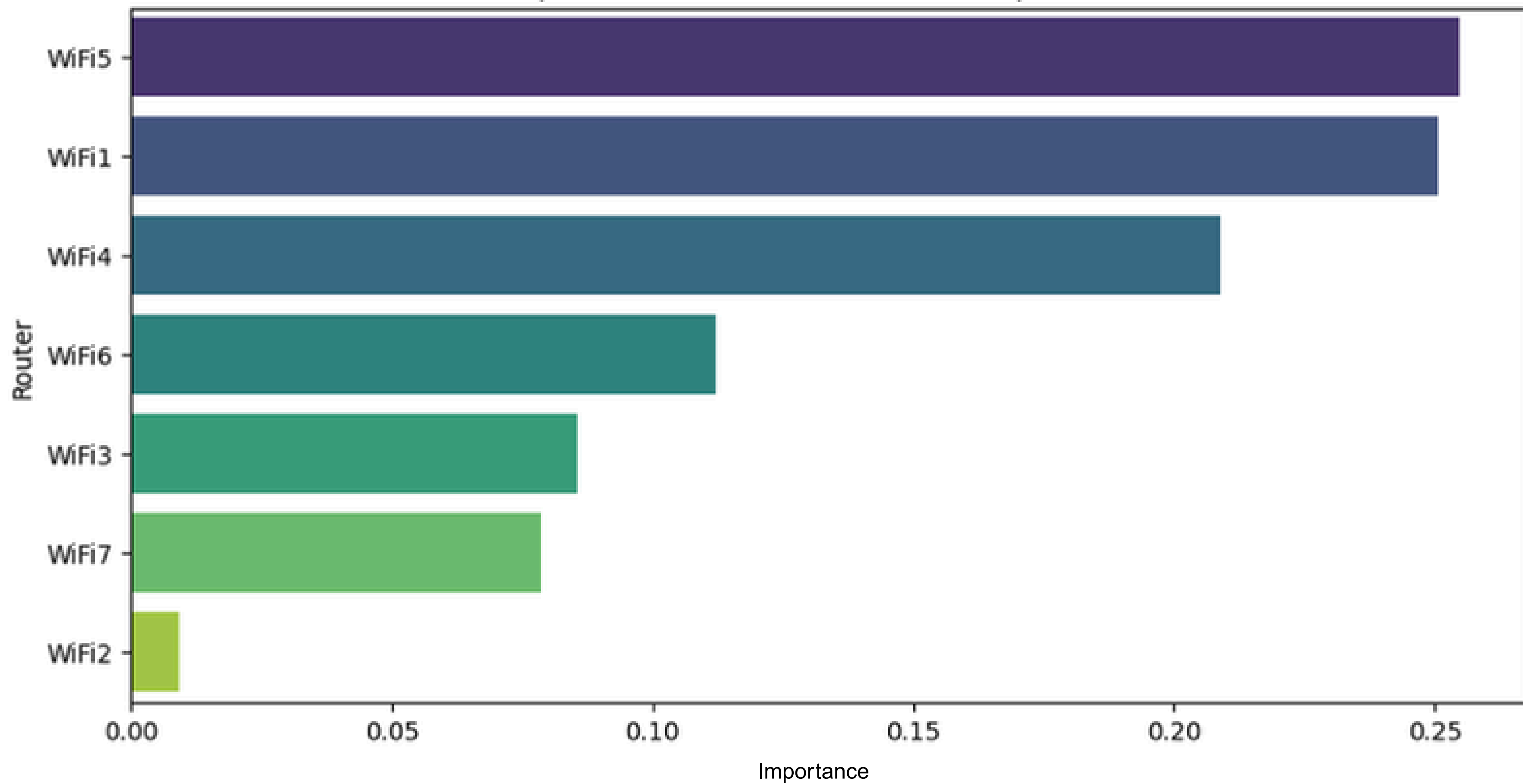


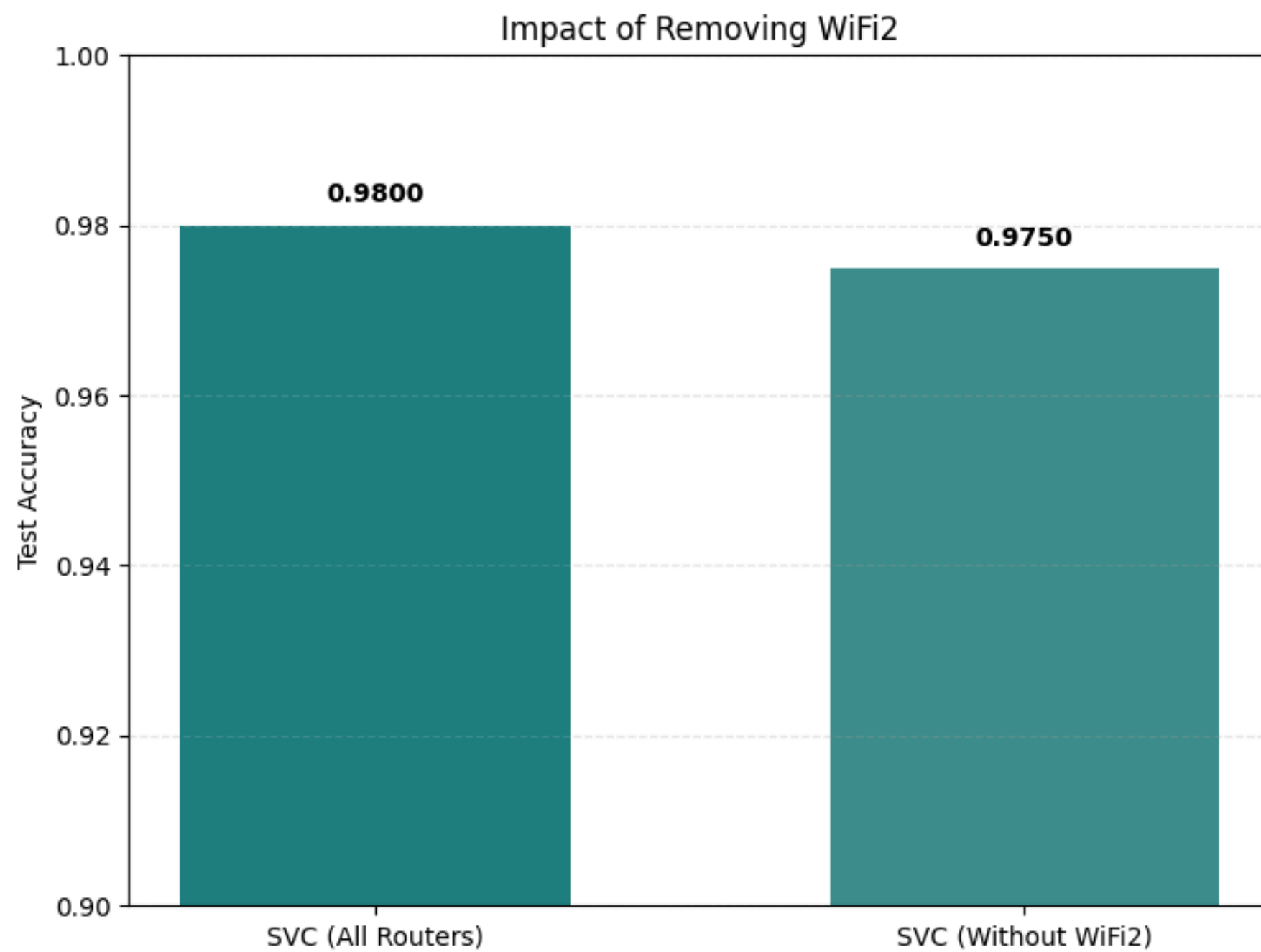
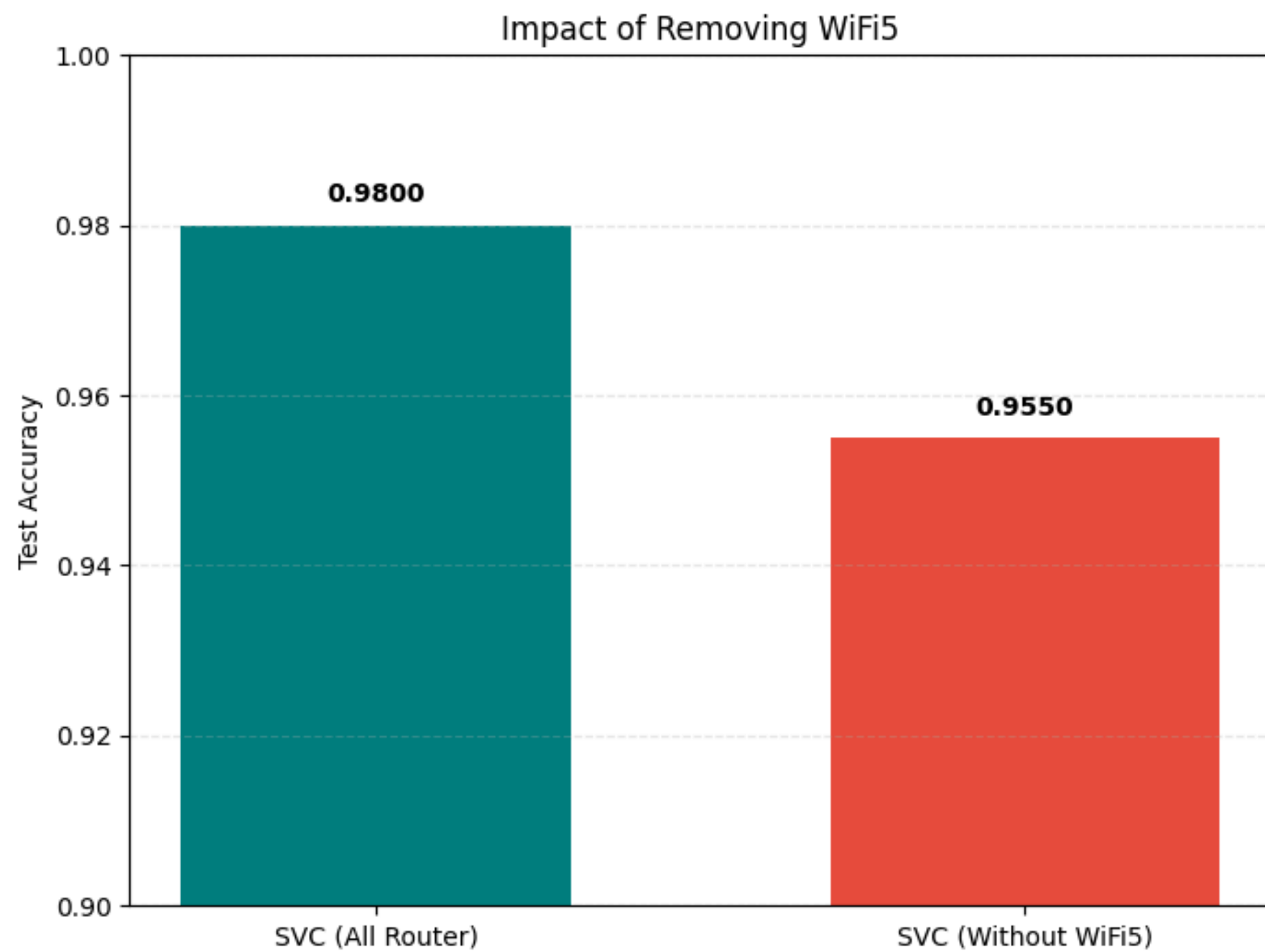
Tabella Riassuntiva dei Risultati - WiFi Localization

Modello	Train Acc	Test Acc	Gap (Overfit)	Precision	Recall	F1-Score
SVC 🏰	0.9857	0.98	0.0057	0.9804	0.98	0.98
Naive Bayes	0.985	0.98	0.005	0.9809	0.98	0.98
Random Forest	1.0	0.9767	0.0233	0.9769	0.9767	0.9766
KNN	0.9921	0.975	0.0171	0.9759	0.975	0.975
Decision Tree	0.9857	0.9717	0.014	0.9721	0.9717	0.9717

Further analysis

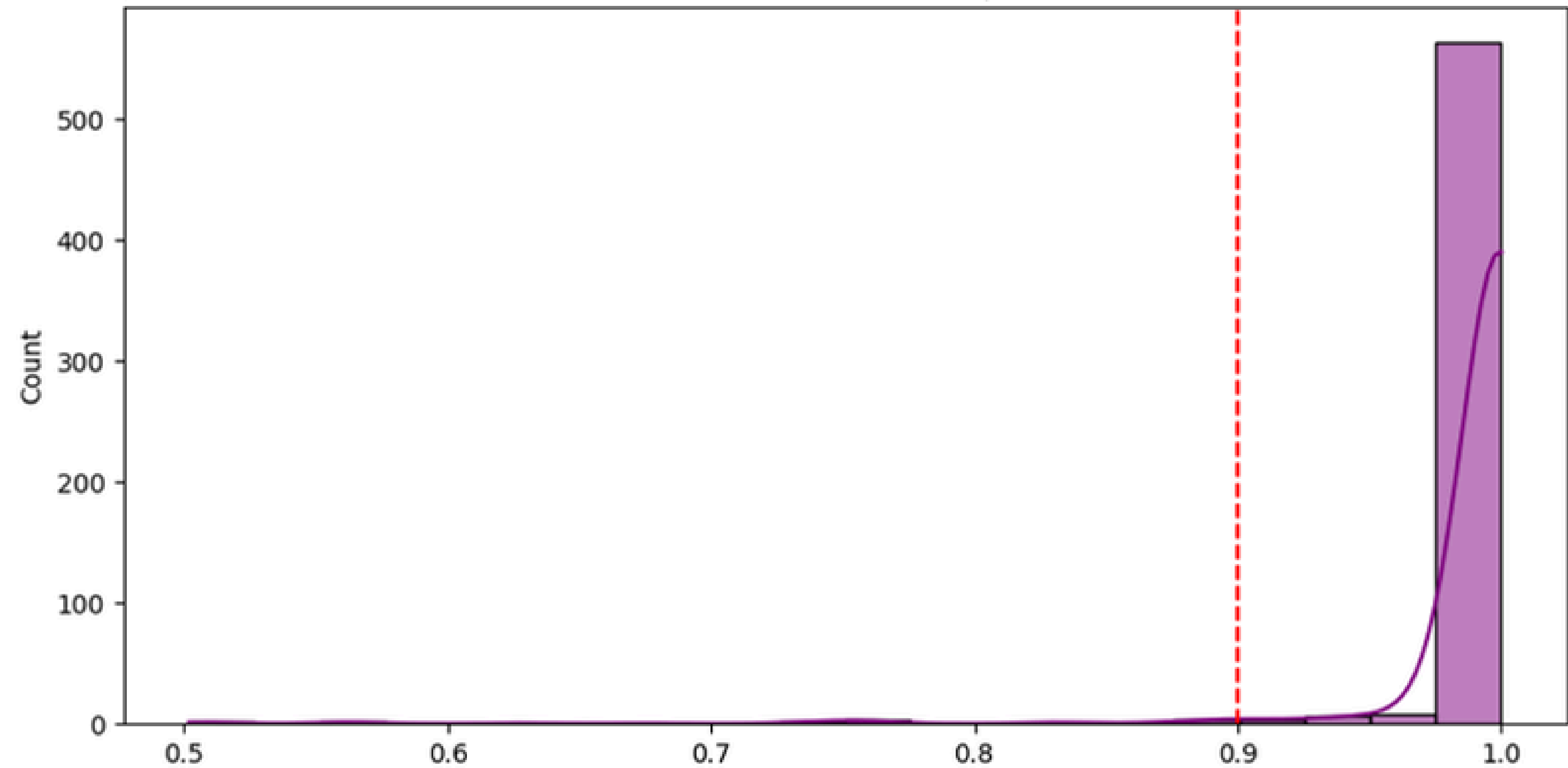
Importance of Routers (Feature Importance)





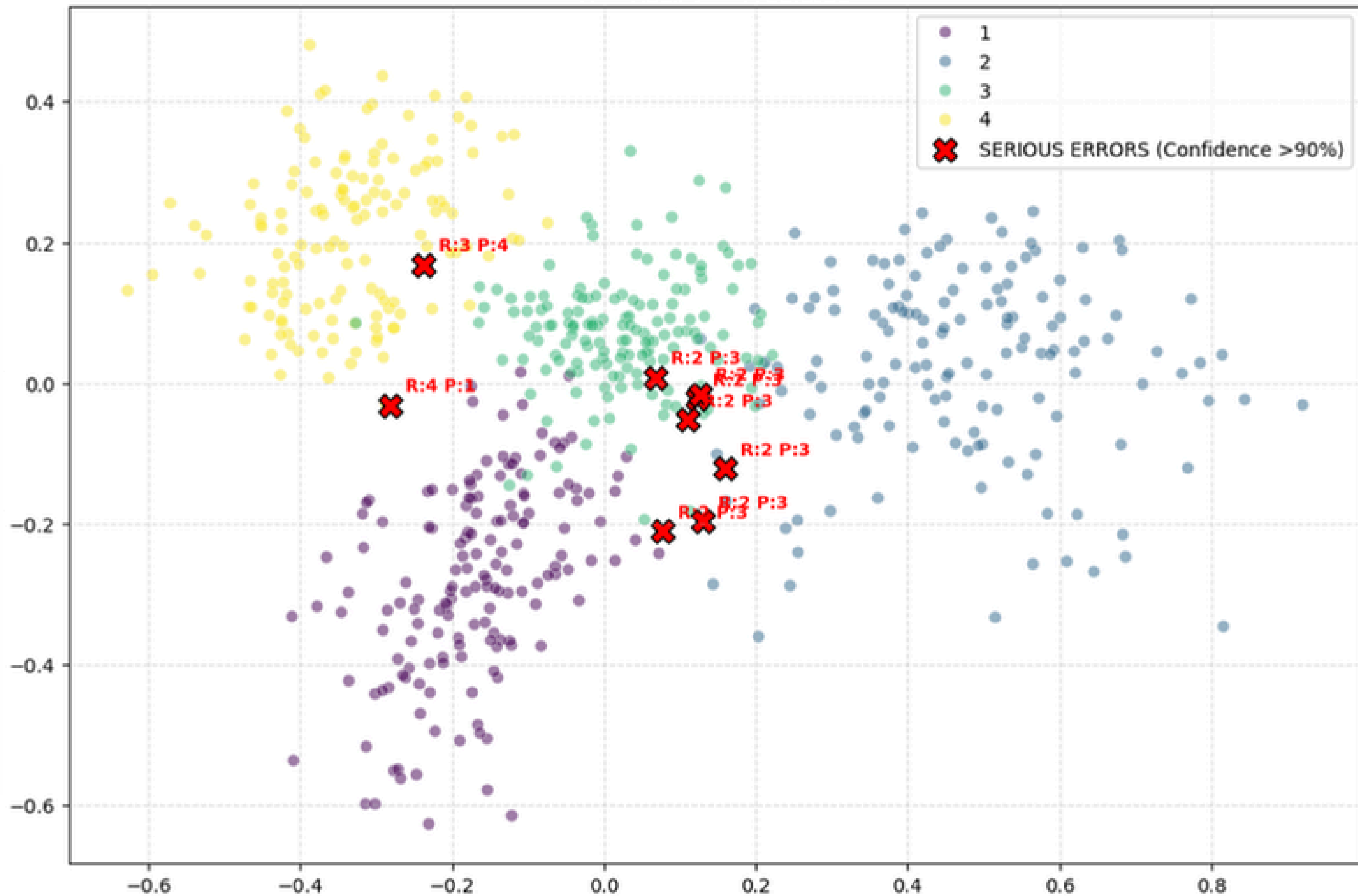
Naive Bayes

Distribution of confidence in predictions



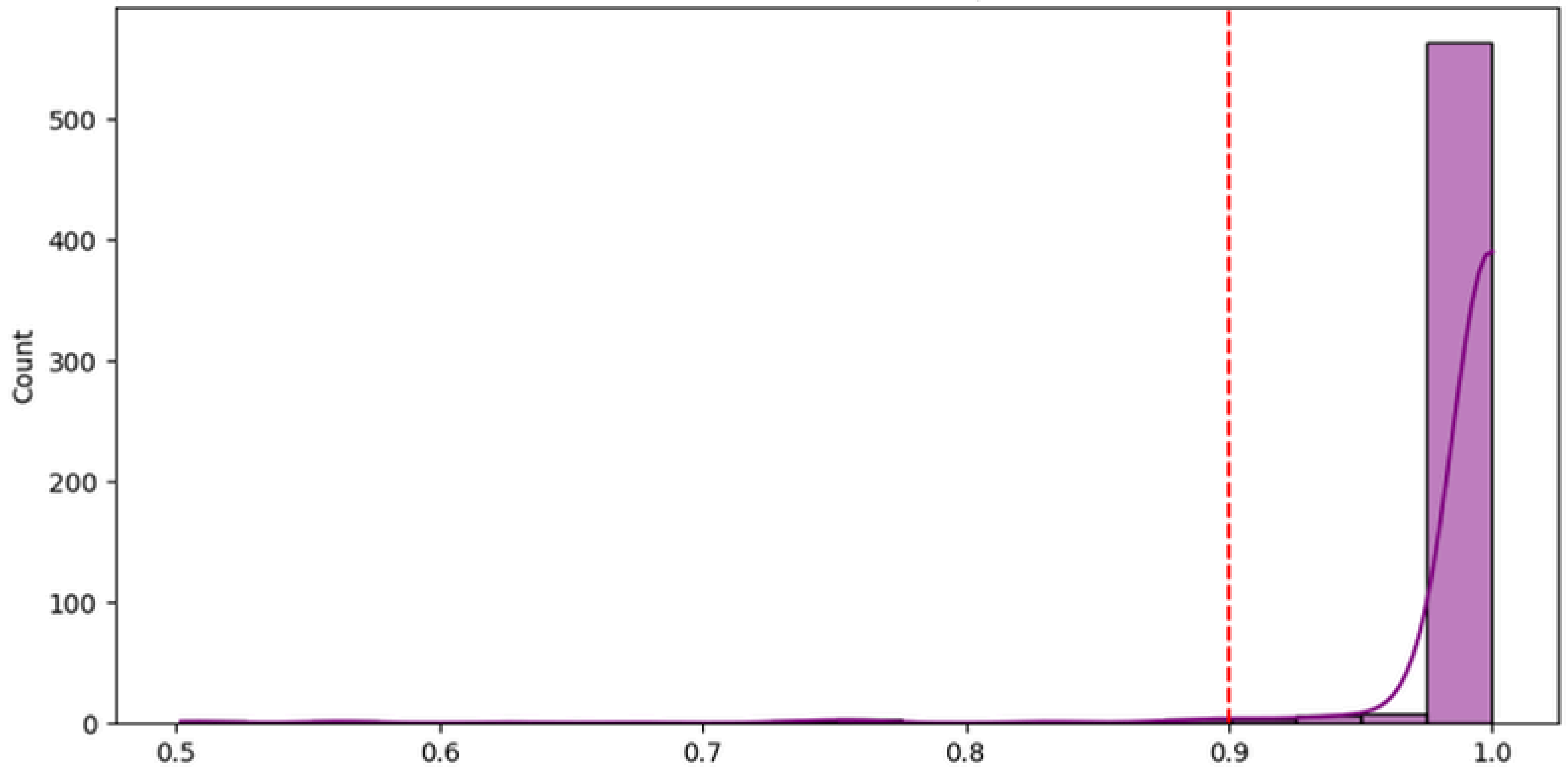
Where do Serious Errors occur? (Naive Bayes)

R = Real, P = Predicted



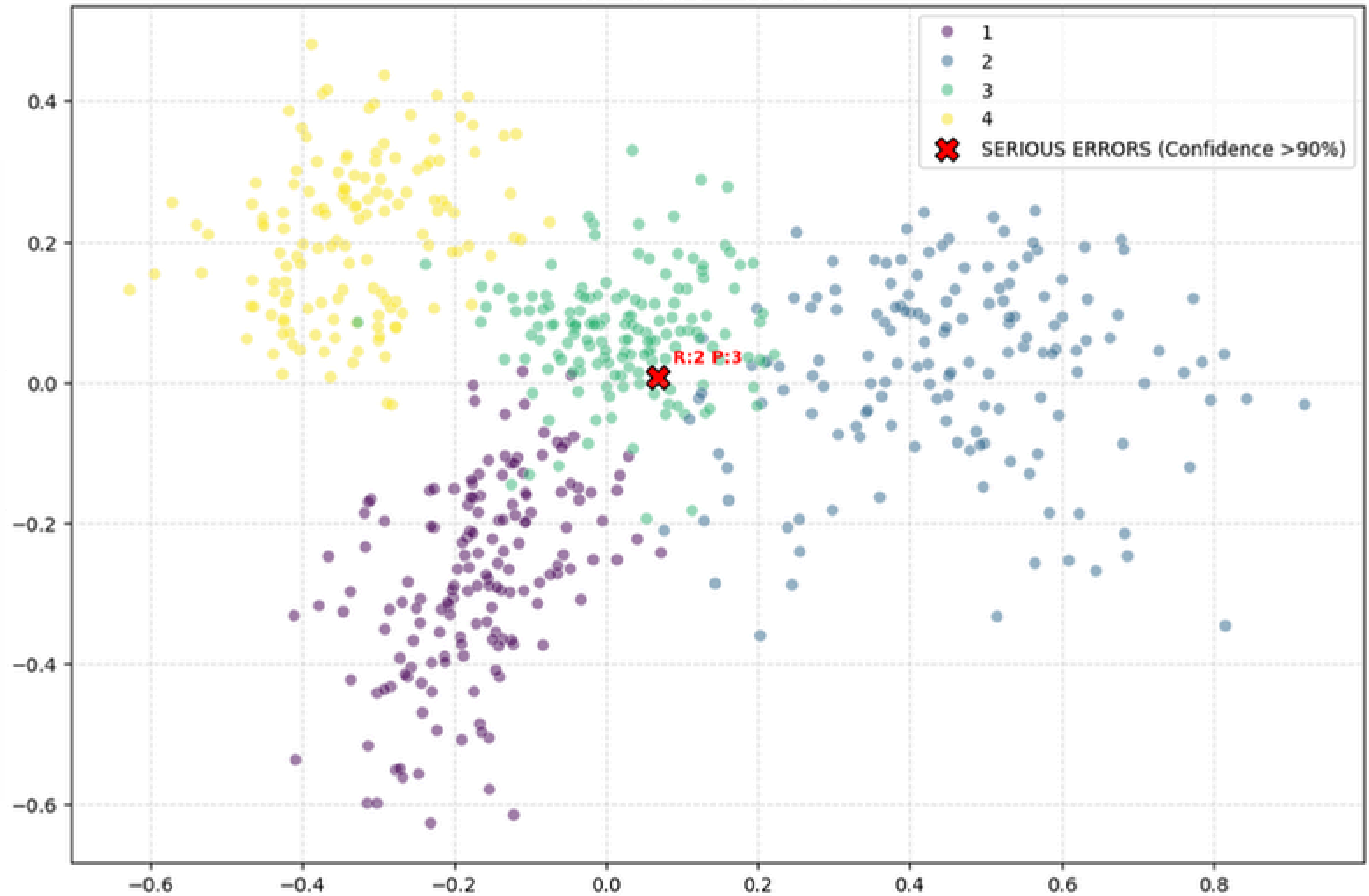
SVC

Distribution of confidence in predictions



Where do Serious Errors occur? (SVC)

R = Real, P = Predicted



**Thank you for your
attention**
